EXPERIMENT - 8
16-BIT DIVISION

AIM: To write an assembly language program to implement 16-bit division using 8085 processor.

ALGORITHM:

1. Read divided (16 bit)
2. Read divisor
3. count ← 8
4. Left shift dividend
5. subtract divisor from upper 8-bits of dividend.
6. If CS = 1 go to 9
7. Restore dividend
8. Increment lower 8-bits of dividend
9. count ← count - 1
10. If count = 0 go to 5
11. Store upper 8-bit dividend as remainder and lower 8-bit as quotient.
12. Stop.

PROGRAM:

```
LDA 8501 — Load dividend
MOV B, A - MOV A to B
LDA 8500 - Load divisor
MVI C,00 - clear c register
LOOP: CMP B - Compare A with B
JC LOOP 1 — Jump if carry
SUB B - Subtract B from A
INR C - Increment C
JMP LOOP - Repeat the loop.
STA 8503 - Store remainder
DCR C - Decrement C
MOV A, C - Move C to A
LOOP 1: STA 8502 - store quotient
RST 1 - Software Interrupt.
```

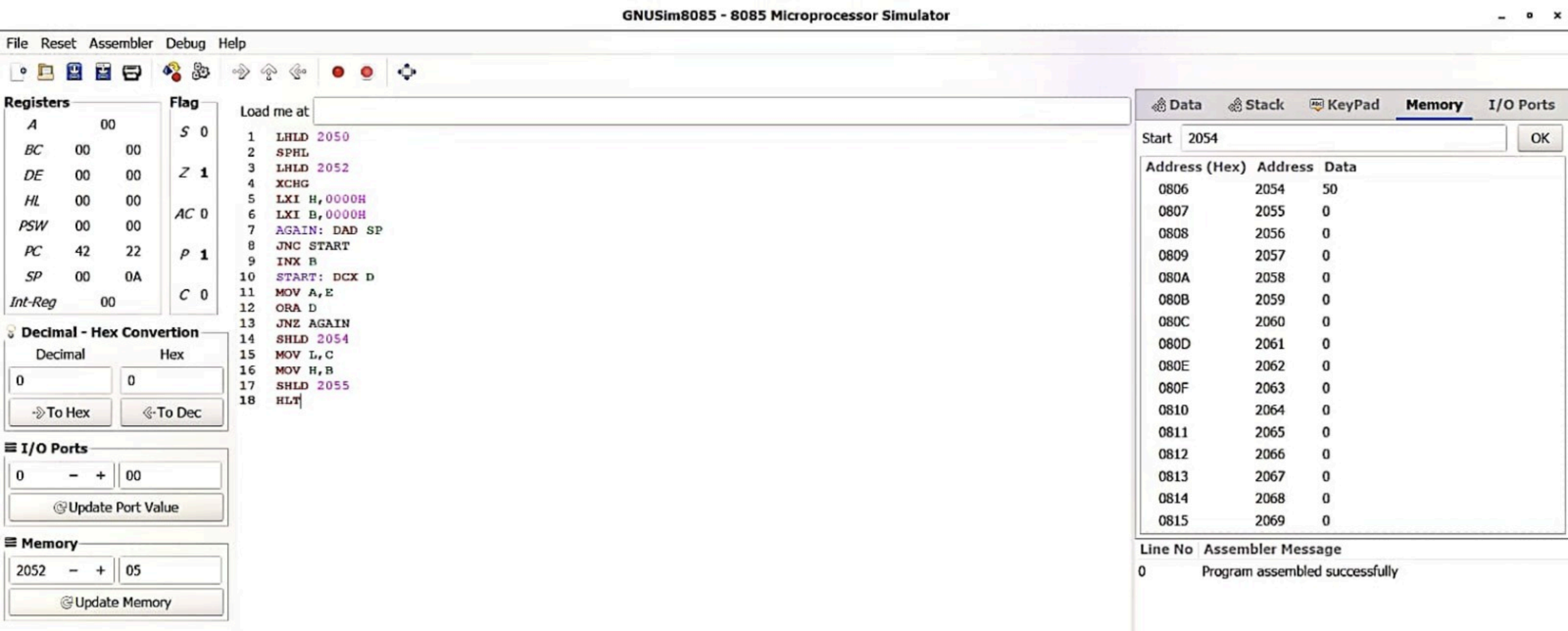
Input:

Address	Data
8051	20
8050	2

Output:

Address	Data
8502	10
8503	2

Result: Thus the program was executed successfully using 8085 processor simulator.


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 00
BC 00 00
DE 00 00
HL 00 00
PSW 00 00
PC 42 22
SP 00 0A
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Decimal - Hex Convertion
Decimal Hex
0 0
To Hex To Dec
I/O Ports
0 00
Update Port Value
Memory
2052 05
Update Memory
Load me at
1 LHLD 2050
2 SPHL
3 LHLD 2052
4 XCHG
5 LXI H,0000H
6 LXI B,0000H
7 AGAIN: DAD SP
8 JNC START
9 INX B
10 START: DCX D
11 MOV A,E
12 ORA D
13 JNZ AGAIN
14 SHLD 2054
15 MOV L,C
16 MOV H,B
17 SHLD 2055
18 HLT
Data Stack KeyPad Memory I/O Ports
Start 2054 OK
Address (Hex) Address Data
0806 2054 50
0807 2055 0
0808 2056 0
0809 2057 0
080A 2058 0
080B 2059 0
080C 2060 0
080D 2061 0
080E 2062 0
080F 2063 0
0810 2064 0
0811 2065 0
0812 2066 0
0813 2067 0
0814 2068 0
0815 2069 0
Line No Assembler Message
0 Program assembled successfully